The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Observer-Relevant Sub-Pre-Core Exact-Shell Transport-Image Core
Collapse to an Observer-Relevant Sub-Pre-Core Exact-Shell Transport-Section Core

Abstract

The current primitive-reservoir observer-reduction frontier already sharpened the live burden inside the exact-shell-transport-core-generating substrate sub-pre-core package: the full sub-pre-core package was compressed to the observer-relevant sub-pre-core exact-shell transport-image core. The next honest move below that theorem is therefore not another same-output deeper-origin relay that leaves the image core unchanged. The sharper benchmark-facing move is to ask whether the image core itself still carries presentational redundancy that can be collapsed away without changing the benchmark one-metric package.

The present note proves that it does. Relative to the already fixed observer-relevant pre-core exact-shell transport core, the only independent sub-pre-core datum still carried by the current image core is the canonical transport section from the recorded pre-core transport body into its sub-pre-core minimizer image. The transport-image body is exactly the image of that section, the restricted projection is its inverse, and the exact-shell transport image is exactly the shell restriction of the same section. Separate listing of those image-core constituents therefore does not add independent benchmark-facing freedom.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion language is licensed. The positive result is therefore a genuine smaller collapse strictly inside the current image core: the live burden now contracts to an observer-relevant sub-pre-core exact-shell transport-section core. What remains open is to derive or justify that section core from still more primitive substrate structure, or else prove a still sharper collapse strictly inside it.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity is again constrained by a routing safeguard. Once a current benchmark-facing datum has been compressed to a smaller observer-relevant core, another theorem that merely replays the same datum through a deeper relay does not count as new front-line progress unless it changes benchmark-facing status by more than depth alone. The sub-pre-core image-core theorem made exactly such a compression: it removed the full exact-shell-transport-core-generating sub-pre-core package as the live burden and replaced it by a smaller observer-relevant sub-pre-core exact-shell transport-image core.

That theorem also exposes the next honest question. Inside the image core itself, what is genuinely independent observer-relevant data, and what is merely presentational bookkeeping once the recorded pre-core exact-shell transport core is already fixed? The present note takes the sharper collapse route rather than a same-output deeper-origin relay. It shows that the image-core body,

shell image, and restricted projection are not independent pieces at current scope. They are all recovered from a single canonical transport section over the already fixed pre-core transport body.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not introduce a second operative metric, it does not enlarge the low-energy matter law, and it does not reopen the already-screened larger sub-pre-core or pre-core packages as if they were again the live burden. Its narrower benchmark-facing role is to isolate the only remaining observer-relevant datum still carried by the current sub-pre-core image core.

2. Fixed Inputs

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}},$$

a visible reduction map

$$\mathcal{R}_\sigma : \mathcal{I}_\sigma^{\mathrm{for}} \to X_\sigma,$$

and shell-level current and equilibrium carriers

$$\mathfrak{C}_\sigma^{\mathrm{cur}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{cur}}, \qquad \mathfrak{C}_\sigma^{\mathrm{eq}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{eq}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$*.*

Definition 2 (Recorded proto-germ exact-shell target)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The active line already fixes:*

1. *the same substrate target and pairing*

$$E_\sigma^{\mathrm{sub}}, \qquad \mathfrak{s}_\sigma^{\mathrm{sub}} : E_\sigma^{\mathrm{sub}} \times E_\sigma^{\mathrm{sub}} \to \mathbb{R};$$

2. *a hidden fiber space* $\mathcal{H}_\sigma^{\mathrm{hid}}$ *with compact hidden body* $D_\sigma^{\mathrm{hid}} \subset \mathcal{H}_\sigma^{\mathrm{hid}}$*;*

3. *a finite-dimensional proto-germ state space* $\mathcal{U}_\sigma^{\mathrm{proto}}$*, a compact convex proto-germ body* $\mathcal{B}_\sigma^{\mathrm{proto}} \subset \mathcal{U}_\sigma^{\mathrm{proto}}$*, a smooth proto-germ restriction section*

$$\Gamma_\sigma^{\mathrm{proto}} : \mathcal{U}_\sigma^{\mathrm{proto}} \to E_\sigma^{\mathrm{sub}},$$

and the exact proto-germ shell

$$\mathcal{Z}_\sigma^{\mathrm{proto}} \equiv \left(\Gamma_\sigma^{\mathrm{proto}}\right)^{-1}(0) \cap \mathcal{B}_\sigma^{\mathrm{proto}};$$

4. *a closed embedded proto zero core* $\mathcal{Z}_\sigma^{0,\mathrm{proto}} \subset \mathcal{Z}_\sigma^{\mathrm{proto}}$ *and a split diffeomorphism*

$$\Phi_\sigma^{\mathrm{proto}} : \mathcal{Z}_\sigma^{0,\mathrm{proto}} \times D_\sigma^{\mathrm{hid}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{proto}};$$

5. *a hidden charge*

$$\mathcal{Q}_\sigma^{\mathrm{proto}} : \mathcal{H}_\sigma^{\mathrm{hid}} \to Q_\sigma^{\mathrm{eqn}},$$

a smooth observer-compatible exact proto-germ zero-response realization

$$\zeta_\sigma^{\mathrm{proto}} : \mathcal{I}_\sigma^{\mathrm{for}} \hookrightarrow \mathcal{Z}_\sigma^{0,\mathrm{proto}},$$

and a fixed-data solvability / coercive package on $\mathcal{Z}_\sigma^{0,\mathrm{proto}}$*, recorded through* $\widetilde{K}_\sigma^{0,\mathrm{proto}}$ *and* $P_\sigma^{\mathrm{proto,co}}$*.*

Definition 3 (Recorded current transport nucleus)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The recorded current transport nucleus* $\mathfrak{Q}^{\mathrm{nuc}}$ *for sector* σ *consists of:*

1. *the pre-proto-germ state space* $\mathcal{V}_\sigma^{\mathrm{preproto}}$*, compact convex body* $\mathcal{B}_\sigma^{\mathrm{preproto}} \subset \mathcal{V}_\sigma^{\mathrm{preproto}}$*, projection*

$$\Pi_\sigma^{\mathrm{preproto}} : \mathcal{V}_\sigma^{\mathrm{preproto}} \to \mathcal{U}_\sigma^{\mathrm{proto}},$$

and proto section

$$\mathfrak{f}_\sigma^{\mathrm{preproto,proto}} : \mathcal{B}_\sigma^{\mathrm{proto}} \to \mathcal{B}_\sigma^{\mathrm{preproto}}$$

satisfying

$$\Pi_\sigma^{\mathrm{preproto}}\left(\mathcal{B}_\sigma^{\mathrm{preproto}}\right) = \mathcal{B}_\sigma^{\mathrm{proto}}, \qquad \Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,proto}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{proto}}};$$

2. *the restriction section*

$$\Gamma_\sigma^{\mathrm{preproto}} : \mathcal{V}_\sigma^{\mathrm{preproto}} \to E_\sigma^{\mathrm{sub}},$$

the induced functional

$$\mathscr{W}_\sigma^{\mathrm{preproto}}(q) \equiv \mathfrak{s}_\sigma^{\mathrm{sub}}\left(\Gamma_\sigma^{\mathrm{preproto}}(q), \Gamma_\sigma^{\mathrm{preproto}}(q)\right),$$

and the exact shell

$$\mathcal{Z}_\sigma^{\mathrm{preproto}} \equiv \left(\Gamma_\sigma^{\mathrm{preproto}}\right)^{-1}(0) \cap \mathcal{B}_\sigma^{\mathrm{preproto}},$$

with

$$\Gamma_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,proto}} = \Gamma_\sigma^{\mathrm{proto}}, \qquad \Pi_\sigma^{\mathrm{preproto}}\big|_{\mathcal{Z}_\sigma^{\mathrm{preproto}}} : \mathcal{Z}_\sigma^{\mathrm{preproto}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{proto}};$$

3. *benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, and no premature promotion from adoption to completed derivation.*

Definition 4 (Recorded observer-relevant pre-nucleus exact-shell core)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The observer-relevant pre-nucleus exact-shell core* $\mathfrak{R}^{\mathrm{core}}$ *for sector* σ *consists of:*

1. *the pre-nucleus state space* $\mathcal{W}_\sigma^{\mathrm{prenuc}}$*, compact convex body* $\mathcal{B}_\sigma^{\mathrm{prenuc}} \subset \mathcal{W}_\sigma^{\mathrm{prenuc}}$*, affine surjection*

$$\Pi_\sigma^{\mathrm{prenuc}} : \mathcal{W}_\sigma^{\mathrm{prenuc}} \to \mathcal{V}_\sigma^{\mathrm{preproto}},$$

and fiber minimizer

$$\mathfrak{f}_\sigma^{\mathrm{prenuc,nuc}} : \mathcal{B}_\sigma^{\mathrm{preproto}} \to \mathcal{B}_\sigma^{\mathrm{prenuc}}$$

satisfying

$$\Pi_\sigma^{\mathrm{prenuc}}(\mathcal{B}_\sigma^{\mathrm{prenuc}}) = \mathcal{B}_\sigma^{\mathrm{preproto}}, \qquad \Pi_\sigma^{\mathrm{prenuc}} \circ \mathfrak{f}_\sigma^{\mathrm{prenuc,nuc}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{preproto}}};$$

2. *a closed embedded exact pre-nucleus shell*

$$\mathcal{Z}_\sigma^{\mathrm{prenuc}} \subset \mathcal{B}_\sigma^{\mathrm{prenuc}}$$

satisfying

$$\mathfrak{f}_\sigma^{\mathrm{prenuc,nuc}}(\mathcal{Z}_\sigma^{\mathrm{preproto}}) \subset \mathcal{Z}_\sigma^{\mathrm{prenuc}}, \qquad \Pi_\sigma^{\mathrm{prenuc}}\big|_{\mathcal{Z}_\sigma^{\mathrm{prenuc}}} : \mathcal{Z}_\sigma^{\mathrm{prenuc}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{preproto}};$$

3. *benchmark faithfulness in the sense of Definition 3.*

Definition 5 (Recorded pre-core exact-shell transport core)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The recorded observer-relevant pre-core exact-shell transport core* $\mathfrak{S}^{\mathrm{tr}}$ *for sector* σ *consists of:*

1. *a compact transport-core body*

$$\mathcal{B}_\sigma^{\mathrm{precore,tr}},$$

an affine projection

$$\widehat{\Pi}_\sigma^{\mathrm{precore}} : \mathcal{B}_\sigma^{\mathrm{precore,tr}} \to \mathcal{B}_\sigma^{\mathrm{prenuc}},$$

and a smooth transport-core lift

$$\mathfrak{f}_\sigma^{\mathrm{precore,tr}} : \mathcal{B}_\sigma^{\mathrm{prenuc}} \to \mathcal{B}_\sigma^{\mathrm{precore,tr}}$$

satisfying

$$\widehat{\Pi}_\sigma^{\mathrm{precore}}(\mathcal{B}_\sigma^{\mathrm{precore,tr}}) = \mathcal{B}_\sigma^{\mathrm{prenuc}}, \qquad \widehat{\Pi}_\sigma^{\mathrm{precore}} \circ \mathfrak{f}_\sigma^{\mathrm{precore,tr}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{prenuc}}};$$

2. *a closed embedded exact transport shell*

$$\mathcal{Z}_\sigma^{\mathrm{precore,tr}} \subset \mathcal{B}_\sigma^{\mathrm{precore,tr}}$$

satisfying

$$\mathfrak{f}_\sigma^{\mathrm{precore,tr}}(\mathcal{Z}_\sigma^{\mathrm{prenuc}}) \subset \mathcal{Z}_\sigma^{\mathrm{precore,tr}}, \qquad \widehat{\Pi}_\sigma^{\mathrm{precore}}\big|_{\mathcal{Z}_\sigma^{\mathrm{precore,tr}}} : \mathcal{Z}_\sigma^{\mathrm{precore,tr}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{prenuc}};$$

3. *benchmark faithfulness in the sense of Definition 4.*

Remark 1. *Definitions 1–5 are fixed inputs. The one operative metric, the ordinary matter route, the fixed proto-germ exact-shell target, the current transport nucleus, the recorded pre-nucleus exact-shell core, and the recorded pre-core exact-shell transport core are not reopened here. The present note asks only which part of the current sub-pre-core image-core presentation still carries independent benchmark-facing burden once those downstream data are already fixed.*

3. Full Sub-Pre-Core Package

Definition 6 (Exact-shell-transport-core-generating substrate sub-pre-core dynamics)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. An exact-shell-transport-core-generating substrate sub-pre-core dynamics package* $\mathfrak{U}^{\mathrm{subpre}}$ *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean sub-pre-core state space* $\mathcal{L}_\sigma^{\mathrm{subpre}}$*, a compact convex sub-pre-core body*

$$\mathcal{B}_\sigma^{\mathrm{subpre}} \subset \mathcal{L}_\sigma^{\mathrm{subpre}}$$

with nonempty interior, an affine surjection

$$\Pi_\sigma^{\mathrm{subpre}} : \mathcal{L}_\sigma^{\mathrm{subpre}} \to \mathcal{B}_\sigma^{\mathrm{precore,tr}},$$

and a smooth sub-pre-core minimizer

$$\mathfrak{f}_\sigma^{\mathrm{subpre,tr}} : \mathcal{B}_\sigma^{\mathrm{precore,tr}} \to \mathcal{B}_\sigma^{\mathrm{subpre}}$$

satisfying

$$\Pi_\sigma^{\mathrm{subpre}}(\mathcal{B}_\sigma^{\mathrm{subpre}}) = \mathcal{B}_\sigma^{\mathrm{precore,tr}}, \qquad \Pi_\sigma^{\mathrm{subpre}} \circ \mathfrak{f}_\sigma^{\mathrm{subpre,tr}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{precore,tr}}}.$$

2. *A closed embedded exact sub-pre-core shell*

$$\mathcal{Z}_\sigma^{\mathrm{subpre}} \subset \mathcal{B}_\sigma^{\mathrm{subpre}}$$

satisfying

$$\mathfrak{f}_\sigma^{\mathrm{subpre,tr}}(\mathcal{Z}_\sigma^{\mathrm{precore,tr}}) \subset \mathcal{Z}_\sigma^{\mathrm{subpre}}, \qquad \Pi_\sigma^{\mathrm{subpre}}\big|_{\mathcal{Z}_\sigma^{\mathrm{subpre}}} : \mathcal{Z}_\sigma^{\mathrm{subpre}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{precore,tr}}.$$

3. *Benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, and no premature promotion from adoption to completed derivation.*

Remark 2. *Definition 6 is the full current sub-pre-core package. The previous active-primary theorem already showed that the full package is not the minimal benchmark-facing datum once the recorded pre-core transport core is fixed.*

4. Observer-Relevant Sub-Pre-Core Exact-Shell Transport-Image Core

Definition 7 (Observer-relevant sub-pre-core exact-shell transport-image core)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ *and a benchmark-faithful sub-pre-core package in the sense of Definition 6. The* observer-relevant sub-pre-core exact-shell transport-image core $\mathfrak{U}^{\mathrm{img}}$ *for sector* σ *consists of:*

1. *the compact transport-image body*

$$\mathcal{B}_\sigma^{\mathrm{subpre,img}} \equiv \mathfrak{f}_\sigma^{\mathrm{subpre,tr}}(\mathcal{B}_\sigma^{\mathrm{precore,tr}}) \subset \mathcal{B}_\sigma^{\mathrm{subpre}},$$

   together with the restricted projection

$$\widehat{\Pi}_\sigma^{\mathrm{subpre}} \equiv \Pi_\sigma^{\mathrm{subpre}}\big|_{\mathcal{B}_\sigma^{\mathrm{subpre,img}}} : \mathcal{B}_\sigma^{\mathrm{subpre,img}} \to \mathcal{B}_\sigma^{\mathrm{precore,tr}};$$

2. *the exact-shell transport image*

$$\mathcal{Z}_\sigma^{\mathrm{subpre,img}} \equiv \mathfrak{f}_\sigma^{\mathrm{subpre,tr}}(\mathcal{Z}_\sigma^{\mathrm{precore,tr}}) \subset \mathcal{Z}_\sigma^{\mathrm{subpre}},$$

   with the restricted shell projection

$$\widehat{\Pi}_\sigma^{\mathrm{subpre}}\big|_{\mathcal{Z}_\sigma^{\mathrm{subpre,img}}} : \mathcal{Z}_\sigma^{\mathrm{subpre,img}} \to \mathcal{Z}_\sigma^{\mathrm{precore,tr}};$$

3. *benchmark faithfulness in the sense of Definition 6.*

Remark 3. *Definition 7 is already smaller than the full sub-pre-core package. The present note asks whether it is itself still larger than necessary once the recorded pre-core exact-shell transport body and shell are treated as fixed upstream data.*

5. Observer-Relevant Sub-Pre-Core Exact-Shell Transport-Section Core

Definition 8 (Observer-relevant sub-pre-core exact-shell transport-section core)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ *and a benchmark-faithful sub-pre-core package in the sense of Definition 6. The* observer-relevant sub-pre-core exact-shell transport-section core $\mathfrak{U}^{\mathrm{sec}}$ *for sector* σ *consists of:*

1. *the canonical transport section*

$$\mathfrak{s}_\sigma^{\mathrm{subpre,sec}} \equiv \mathfrak{f}_\sigma^{\mathrm{subpre,tr}} : \mathcal{B}_\sigma^{\mathrm{precore,tr}} \to \mathcal{B}_\sigma^{\mathrm{subpre}};$$

2. *benchmark faithfulness in the sense of Definition 6.*

The shell restriction

$$\mathfrak{s}_\sigma^{\mathrm{subpre,sec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{precore,tr}}} : \mathcal{Z}_\sigma^{\mathrm{precore,tr}} \to \mathcal{Z}_\sigma^{\mathrm{subpre}}$$

is understood as derived data rather than separately listed core data.

Remark 4. *Definition 8 is strictly smaller than Definition 7 at current scope. Once the recorded transport-core body* $\mathcal{B}_\sigma^{\mathrm{precore,tr}}$ *and shell* $\mathcal{Z}_\sigma^{\mathrm{precore,tr}}$ *are fixed, the image body, image shell, and restricted projection are recovered from the single canonical section* $\mathfrak{s}_\sigma^{\mathrm{subpre,sec}}$*; they need not be listed independently.*

6. Collapse of the Image Core

Proposition 1 (The canonical transport section reconstructs the image body and projection)**.** *For every benchmark-faithful sub-pre-core package,*

$$\mathcal{B}_\sigma^{\mathrm{subpre,img}} = \mathfrak{s}_\sigma^{\mathrm{subpre,sec}}(\mathcal{B}_\sigma^{\mathrm{precore,tr}}),$$

and the restricted projection

$$\widehat{\Pi}_\sigma^{\mathrm{subpre}} : \mathcal{B}_\sigma^{\mathrm{subpre,img}} \to \mathcal{B}_\sigma^{\mathrm{precore,tr}}$$

is a homeomorphism with inverse $\mathfrak{s}_\sigma^{\mathrm{subpre,sec}}$*.*

Proof. The displayed identity is Definition 7(1) together with Definition 8(1). By Definition 6(1),

$$\Pi_\sigma^{\mathrm{subpre}} \circ \mathfrak{s}_\sigma^{\mathrm{subpre,sec}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{precore,tr}}}.$$

Restricting $\Pi_\sigma^{\mathrm{subpre}}$ to $\mathcal{B}_\sigma^{\mathrm{subpre,img}}$ yields

$$\widehat{\Pi}_\sigma^{\mathrm{subpre}} \circ \mathfrak{s}_\sigma^{\mathrm{subpre,sec}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{precore,tr}}}.$$

Conversely, every $y \in \mathcal{B}_\sigma^{\mathrm{subpre,img}}$ has the form

$$y = \mathfrak{s}_\sigma^{\mathrm{subpre,sec}}(w) \qquad \text{for some } w \in \mathcal{B}_\sigma^{\mathrm{precore,tr}}.$$

Hence

$$\begin{aligned}
\mathfrak{s}_\sigma^{\mathrm{subpre,sec}}(\widehat{\Pi}_\sigma^{\mathrm{subpre}}(y)) &= \mathfrak{s}_\sigma^{\mathrm{subpre,sec}}(\Pi_\sigma^{\mathrm{subpre}}(y)) \\
&= \mathfrak{s}_\sigma^{\mathrm{subpre,sec}}(\Pi_\sigma^{\mathrm{subpre}}(\mathfrak{s}_\sigma^{\mathrm{subpre,sec}}(w))) \\
&= \mathfrak{s}_\sigma^{\mathrm{subpre,sec}}(w) \\
&= y.
\end{aligned}$$

Therefore $\widehat{\Pi}_\sigma^{\mathrm{subpre}}$ and $\mathfrak{s}_\sigma^{\mathrm{subpre,sec}}$ are mutually inverse continuous maps, so the restricted projection is a homeomorphism with inverse $\mathfrak{s}_\sigma^{\mathrm{subpre,sec}}$. □

Proposition 2 (The shell image is exactly the shell restriction of the canonical section)**.** *For every benchmark-faithful sub-pre-core package,*

$$\mathcal{Z}_\sigma^{\mathrm{subpre,img}} = \mathfrak{s}_\sigma^{\mathrm{subpre,sec}}(\mathcal{Z}_\sigma^{\mathrm{precore,tr}}),$$

and the shell restriction

$$\mathfrak{s}_\sigma^{\mathrm{subpre,sec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{precore,tr}}} : \mathcal{Z}_\sigma^{\mathrm{precore,tr}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{subpre,img}}$$

is a diffeomorphism with inverse

$$\widehat{\Pi}_\sigma^{\mathrm{subpre}}\big|_{\mathcal{Z}_\sigma^{\mathrm{subpre,img}}} : \mathcal{Z}_\sigma^{\mathrm{subpre,img}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{precore,tr}}.$$

Proof. The displayed equality is Definition 7(2) together with Definition 8(1). By Definition 6(2),

$$\Pi_\sigma^{\mathrm{subpre}}\big|_{\mathcal{Z}_\sigma^{\mathrm{subpre}}} : \mathcal{Z}_\sigma^{\mathrm{subpre}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{precore,tr}}$$

is a diffeomorphism, and

$$\mathfrak{s}_\sigma^{\mathrm{subpre,sec}}\big(\mathcal{Z}_\sigma^{\mathrm{precore,tr}}\big) \subset \mathcal{Z}_\sigma^{\mathrm{subpre}}.$$

Restricting the shell diffeomorphism to $\mathcal{Z}_\sigma^{\mathrm{subpre,img}}$ gives

$$\widehat{\Pi}_\sigma^{\mathrm{subpre}}\big|_{\mathcal{Z}_\sigma^{\mathrm{subpre,img}}} \circ \mathfrak{s}_\sigma^{\mathrm{subpre,sec}}\big|_{\mathcal{Z}_\sigma^{\mathrm{precore,tr}}} = \mathrm{id}_{\mathcal{Z}_\sigma^{\mathrm{precore,tr}}}.$$

Conversely, if $z \in \mathcal{Z}_\sigma^{\mathrm{subpre,img}}$, then $z = \mathfrak{s}_\sigma^{\mathrm{subpre,sec}}(w)$ for some $w \in \mathcal{Z}_\sigma^{\mathrm{precore,tr}}$, so

$$\mathfrak{s}_\sigma^{\mathrm{subpre,sec}}\big(\widehat{\Pi}_\sigma^{\mathrm{subpre}}(z)\big) = \mathfrak{s}_\sigma^{\mathrm{subpre,sec}}(w) = z.$$

Thus the shell restriction of the canonical section and the restricted shell projection are inverse diffeomorphisms. □

Proposition 3 (The full image core is reconstructed from the section core)**.** *For every benchmark-faithful sub-pre-core package, the full observer-relevant sub-pre-core exact-shell transport-image core* $\mathfrak{U}^{\mathrm{img}}$ *is reconstructed from the smaller section core* $\mathfrak{U}^{\mathrm{sec}}$ *together with the fixed transport-core data* $\mathcal{B}_\sigma^{\mathrm{precore,tr}}$ *and* $\mathcal{Z}_\sigma^{\mathrm{precore,tr}}$*.*

Proof. By Proposition 1, the image body is the image of the canonical section and the restricted projection is its inverse. By Proposition 2, the shell image is the shell restriction of that same section. Benchmark faithfulness is already part of Definition 8(2). Therefore each constituent of Definition 7 is recovered from Definition 8 plus the fixed upstream transport-core body and shell. □

Proposition 4 (Downstream exact-shell transport factors through the section core)**.** *For every benchmark-faithful sub-pre-core package, the recovery of the recorded pre-core exact-shell transport core and all downstream shell transport to the fixed pre-nucleus exact-shell core, transport nucleus, and proto-germ exact-shell target are already carried by the section core* $\mathfrak{U}^{\mathrm{sec}}$ *together with the fixed transport-core data.*

Proof. Proposition 3 reconstructs the full image core from the section core. The recorded transport-core projection

$$\widehat{\Pi}_\sigma^{\mathrm{precore}} : \mathcal{B}_\sigma^{\mathrm{precore,tr}} \to \mathcal{B}_\sigma^{\mathrm{prenuc}}$$

and lift

$$\mathfrak{f}_\sigma^{\mathrm{precore,tr}} : \mathcal{B}_\sigma^{\mathrm{prenuc}} \to \mathcal{B}_\sigma^{\mathrm{precore,tr}}$$

are already fixed by Definition 5. Hence the composite

$$\widehat{\Pi}_\sigma^{\mathrm{precore}} \circ \widehat{\Pi}_\sigma^{\mathrm{subpre}} : \mathcal{B}_\sigma^{\mathrm{subpre,img}} \to \mathcal{B}_\sigma^{\mathrm{prenuc}}$$

and its shell restriction are recovered from the section core because $\widehat{\Pi}_\sigma^{\mathrm{subpre}} = (\mathfrak{s}_\sigma^{\mathrm{subpre,sec}})^{-1}$ on $\mathcal{B}_\sigma^{\mathrm{subpre,img}}$. The same argument then composes with the fixed shell diffeomorphisms of Definitions 4 and 3 to recover downstream transport to $\mathcal{Z}_\sigma^{\mathrm{preproto}}$ and $\mathcal{Z}_\sigma^{\mathrm{proto}}$. Therefore nothing benchmark-facing downstream of the image core requires more than the section core together with the already recorded transport-core package. □

Proposition 5 (Separate image-core presentation is benchmark neutral at current scope)**.** *For every benchmark-faithful sub-pre-core package, the separate presentation of*

$$\mathcal{B}_\sigma^{\mathrm{subpre,img}}, \qquad \widehat{\Pi}_\sigma^{\mathrm{subpre}}, \qquad \mathcal{Z}_\sigma^{\mathrm{subpre,img}}$$

does not add independent benchmark-facing freedom beyond the section core $\mathfrak{U}^{\mathrm{sec}}$*.*

Proof. By Proposition 3, the listed constituents are reconstructed from the canonical transport section together with the already fixed transport-core body and shell. By Proposition 4, all downstream transport relevant to the recorded pre-nucleus exact-shell core, current transport nucleus, and fixed proto-germ exact-shell target is already carried by that reconstructed data. Therefore the image-core presentation as a separate body, shell, and projection is not additional benchmark-facing information beyond the smaller section core. □

Proposition 6 (The benchmark package remains fixed)**.** *Every observer-relevant sub-pre-core exact-shell transport-section core preserves the same benchmark one-metric package* (1)*. In particular, the operative metric remains exactly* $g^\sharp$*, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.*

Proof. This is part of benchmark faithfulness in Definition 8(2). The present note only removes presentational redundancy inside the current image core. It does not alter the benchmark exact-closure package. □

7. Main Theorem

Theorem 1 (Observer-relevant sub-pre-core exact-shell transport-image core collapse to an observer-relevant sub-pre-core exact-shell transport-section core)**.** *Fix the primitive continuation data of Definition 1, the recorded proto-germ exact-shell target of Definition 2, the recorded current transport nucleus of Definition 3, the recorded observer-relevant pre-nucleus exact-shell core of Definition 4, the recorded observer-relevant pre-core exact-shell transport core of Definition 5, and a benchmark-faithful sub-pre-core package in the sense of Definition 6. Then, for each sector* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, the live benchmark-facing burden inside the current sub-pre-core image-core frontier collapses from the observer-relevant sub-pre-core exact-shell transport-image core* $\mathfrak{U}^{\mathrm{img}}$ *to the smaller observer-relevant sub-pre-core exact-shell transport-section core* $\mathfrak{U}^{\mathrm{sec}}$*. More precisely:*

1. *the image body and restricted projection are reconstructed from the canonical transport section by Proposition 1;*
2. *the exact-shell transport image is exactly the shell restriction of that section by Proposition 2;*
3. *the full image core is therefore recovered from the section core and the fixed transport-core body and shell by Proposition 3;*
4. *all downstream recovery of the recorded pre-core exact-shell transport core, the pre-nucleus exact-shell core, the current transport nucleus, and the fixed proto-germ exact-shell target factors through the section core by Proposition 4;*
5. *the separate image-core presentation adds no independent benchmark-facing freedom beyond the section core by Proposition 5;*

6. *the benchmark boundary remains fixed by Proposition 6;*

7. *consequently the remaining honest burden below the previous image-core frontier is no longer the full observer-relevant sub-pre-core exact-shell transport-image core* $\mathfrak{U}^{\mathrm{img}}$ *as such. What remains is to derive or justify the sharper section core* $\mathfrak{U}^{\mathrm{sec}}$ *from still more primitive substrate structure, or else prove a still smaller theorem-level collapse strictly inside that section core.*

Proof. Item (1) is Proposition 1. Item (2) is Proposition 2. Item (3) is Proposition 3. Item (4) is Proposition 4. Item (5) is Proposition 5. Item (6) is Proposition 6. Item (7) is the routing consequence of the previous items together with the active safeguard against counting another same-output deeper-origin relay as new front-line progress when the live benchmark-relevant datum is unchanged. □

Corollary 1 (What must be done next)**.** *After Theorem 1, the next honest benchmark-facing manuscript must either:*

1. *derive or justify the observer-relevant sub-pre-core exact-shell transport-section core* $\mathfrak{U}^{\mathrm{sec}}$ *from still more primitive substrate structure in a way that changes benchmark-facing status;*

2. *identify a genuinely smaller theorem-level observer-relevant datum strictly inside* $\mathfrak{U}^{\mathrm{sec}}$ *and prove a sharper collapse to it;*

3. *do either of the previous two while preserving the same benchmark one-metric package and the same claim boundary.*

Another same-output deeper-origin relay that preserves this section core, another re-expansion back to the larger image-core presentation as if its recovered constituents were still independently live, another reopening of the already-screened full sub-pre-core or pre-core packages as if they were again the active burden, or any move that introduces a second operative metric, enlarges the ordinary matter law, or uses stronger promotion language does not satisfy the present burden. If a quantum continuation is pursued, it matters benchmark-facingly only insofar as it derives this same section core, collapses it further, or closes a benchmark derivation gate in a genuinely new way.

Proof. Theorem 1 shows that the image-core presentation is no longer the minimal benchmark-facing datum at this stage. Therefore a subsequent manuscript must improve on the smaller section core rather than merely restating the larger image core. □

8. Conclusion

The positive result of this note is a disciplined second compression step inside the current sub-pre-core frontier. The previous active-primary theorem already showed that the full sub-pre-core package is not the minimal benchmark-facing datum, because its ambient presentation collapses to an observer-relevant exact-shell transport-image core. The present theorem then takes the honest next internal route available there: it shows that even that image-core presentation is still larger than necessary.

What survives as the live burden is the observer-relevant sub-pre-core exact-shell transport-section core. Relative to the already fixed pre-core exact-shell transport body and shell, that sharper datum is just the canonical transport section that picks out the sub-pre-core minimizer image. The image body is its image, the exact-shell transport image is its shell restriction, and the restricted

projection is simply its inverse. Those separately listed image-core constituents therefore do not add independent benchmark-facing freedom.

This is the honest next burden after the present theorem. A new primary manuscript must derive or justify that sharper section core from still more primitive substrate structure, or else collapse it further, while preserving the same benchmark one-metric package and the same claim boundary. The current result does not yet complete a first-principles derivation of GR from substrate microphysics, but it does narrow the live derivational burden one level further on the active line.